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Day: _____

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Class: CS-F

Home work 35

Problem: Orthogonal Diagonalization
symmetric matrices

$$A = \begin{bmatrix} 2 & 2 & -2 \\ 2 & -1 & 4 \\ -2 & 4 & -1 \end{bmatrix}$$

- Symmetric matrices always give real Eigen values regardless of +ve, -ve, indefinite matrix.
- Eigen vectors corresponding to different eigen values are orthogonal.
- In diagonalization P is orthogonal matrix of normalized eigenvectors, where D is diagonal matrix of eigen values. $A = PDP^T$ $\therefore P^T = P^{-1}$ orthogonal matrix

for Eigen Values

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 2 & -2 \\ 2 & -1-\lambda & 4 \\ -2 & 4 & -1-\lambda \end{vmatrix}$$

$$= (2-\lambda)(-1-\lambda)(-1-\lambda) + (-16) + (-16) - (4(-1-\lambda)) - 16(2-\lambda) - 4(-1-\lambda)$$

$$= -\lambda^3 + 2\lambda^2 + \lambda - 2 - 16 - 16 + 4 + 4\lambda - 32 + 16\lambda + 4 + 4\lambda$$

$$\Rightarrow \lambda^3 - 27\lambda + 54 = 0$$

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$$\lambda_1 = 3, \lambda_2 = -6$$

$$\lambda_2 = -6$$

$$\left[\begin{array}{ccc|c} 8 & 2 & -2 & 0 \\ 2 & 5 & 4 & 0 \\ -2 & 4 & 5 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -1/2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right], x = \begin{bmatrix} 1/2 \\ -1 \\ 1 \end{bmatrix} x_3$$

$$\lambda_1 = 3$$

$$\left[\begin{array}{ccc|c} -1 & 2 & -2 & 0 \\ 2 & -4 & 4 & 0 \\ -2 & 4 & -4 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right], x = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} x_2 + \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} x_3$$

Eigen Vectors

$$\begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

These two vectors are from same eigenvector thus are not orthogonal.

By Gram Schmidt:

$$\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} - (-4) \frac{1}{5} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + \frac{4}{5} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 4/5 \\ 1 \end{bmatrix}$$

vectors are $\begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2/5 \\ 4/5 \\ 1 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & -2/5 \\ -2 & 1 & 4/5 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} -6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2/5 \\ -2 & 1 & 4/5 \\ 2 & 0 & 1 \end{bmatrix}$$